Small Target Detection in UAV Remote Sensing Images Based on Adaptive Cross-Modal Feature Fusion and Attention Mechanism

Zhe Yuan, Baolong Guo, *Member, IEEE*, Jiawei Song, Junbin Zhuang, and Yichuan Ma

***Abstract*—Small target detection in uncrewed aerial vehicle (UAV) remote sensing images faces challenges such as insufficient feature information and background interference in complex environments. Existing single-modal methods, which rely solely on one data source, struggle to address these issues effectively. This article proposes cross-modal feature fusion and attention mechanism YOLO (CA-YOLO), a lightweight detection framework based on adaptive cross-modal feature fusion (ACFF) and an attention mechanism. The framework employs a dual-branch backbone network to extract complementary features from visible (RGB) and infrared (IR) modalities, utilizes an ACFF module to dynamically integrate global semantics and local details, and leverages a lightweight hybrid attention (LHA) mechanism to enhance targets and suppress noise. Experiments on three public datasets, DroneVehicle, low-light visible-infrared paired dataset (LLVIP), and FLIR infrared image super-resolution dataset (FLIR), demonstrate that CA-YOLO achieves mAPs of 75.1%, 96.1%, and 71.4%, respectively, while requiring only 6.8 MB of model parameters and 16.4 giga floating-point operations per second (GFLOPs) of computational cost, significantly outperforming state-of-the-art methods. The inference speeds reach 57.1, 78.1, and 56.5 frames/s on the respective datasets, achieving an optimal balance between accuracy and efficiency. Ablation studies and visualization results further validate the effectiveness of each module and the framework's superior capability in reducing missed detection and false alarm rates (FARs).**

***Index Terms*—Adaptive cross-modal integration, attention mechanism, feature fusion, small target detection, uncrewed aerial vehicles (UAVs).**

I. Introduction

In THE context of rapid technological advancement, uncrewed aerial vehicle (UAV) remote sensing technology has emerged as a critical tool for ground information acquisition, owing to its flexible deployment and high-resolution imaging capabilities. It is widely employed in military reconnaissance, disaster monitoring, environmental protection, and other domains [1], [2]. For example, timely detection of small enemy targets in military reconnaissance can support tactical decision-making, while rapid identification of trapped individuals or damaged infrastructure in disaster scenarios significantly improves rescue efficiency. Consequently, small target detection in UAV remote sensing imagery has become a focal point of research in both academic and industrial communities.

Deep learning has substantially advanced small object detection methodologies [3], [4], [5]. However, challenges remain in complex environments characterized by variable illumination and extreme weather conditions, which affect algorithmic robustness. In low-light conditions, visible (RGB) images suffer from blurred details due to insufficient illumination, while strong light or backlighting often leads to overexposure or underexposure, degrading detection performance. Although infrared (IR) imagery can effectively capture thermal contours of targets under adverse conditions such as darkness or smoke, it lacks color and textural information in well-lit environments. As a result, single-modal imaging approaches exhibit significant limitations in complex real-world scenarios.

To address these limitations, deep learning-based multimodal fusion methods that integrate RGB and IR data have garnered increasing attention. By combining the color and texture details from RGB images with the thermal contour information from IR images, these approaches enhance both detection accuracy and robustness. Lee et al. [6] introduced the cross-modal attention Transformer (CAT) and multimodal fusion Transformer (MFT), which explore modality-specific features and intermodal correlations. Zhou et al. [7] developed the FusionU10 model, which fuses RGB and IR inputs through an enhanced UNet architecture to generate pseudo-color images, achieving strong performance in low-light and occluded environments when combined with YOLOv10 for detection. Sousa et al. [8] extended the YOLOv5 framework by incorporating both early and late fusion strategies, offering practical insights into multimodal object detection.

Multimodal fusion is typically categorized into three paradigms: early fusion, mid-level fusion, and late fusion. Early fusion integrates raw data at the input level and is computationally straightforward, but may lead to loss of modality-specific details. Late fusion combines decisions or outputs from independently processed modalities, preserving feature autonomy but limiting interaction depth. In contrast, mid-level fusion operates during the feature extraction phase, enabling effective integration of complementary features across modalities and has demonstrated superior performance in tasks such as pedestrian detection [9], [10], [11].

Despite notable progress in multimodal fusion for pedestrian detection, its application to small target detection in UAV imagery remains largely exploratory. UAV-captured images present unique challenges, including high-altitude perspectives, large scene coverage, small and sparsely distributed

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The authors are with the Institute of Intelligent Control and Image Engineering, Xidian University, Xi'an 710071, China (e-mail: zheyuan@stu.xidian.edu.cn; blguo@xidian.edu.cn).

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targets, and complex backgrounds—factors that impose stringent demands on fusion techniques. Xu et al. [12] proposed the YOLO-SA method, which enhances key features via adaptive feature aggregation; however, it exhibits limited capability in modeling global contextual information. Wang et al. [13] introduced the CRSIOD network, which extracts shared and complementary features through multiple modules, but its high computational complexity hinders deployment on resource-constrained platforms. Guo et al. [14] developed MMYFNet with a cosine similarity-based fusion module, yet its performance is constrained by the limited receptive field of convolutional neural networks, resulting in suboptimal modeling of long-range dependencies.

In response to the challenges associated with multimodal fusion technology for small target detection in UAV remote sensing images, as well as the limitations in small target feature information, this article presents a detection algorithm that leverages adaptive cross-modal feature fusion (ACFF) and an attention mechanism. The objective is to enhance both the accuracy and robustness of detection by effectively integrating strategies for local and global feature fusion and enhancement. The primary contributions of this article are outlined as follows.

1) A novel multimodal detection network named CA-YOLO is proposed for UAV small target detection. Built upon YOLOv10, the network establishes a dual-branch feature extraction architecture with a parameter-sharing mechanism to synchronously optimize feature representations from RGB and IR modalities.
2) An ACFF module is designed to achieve deep cross-modal integration by dynamically adjusting global semantic information and local detail features between complementary modalities.
3) A lightweight hybrid attention (LHA) module is introduced, employing a lightweight global–local dual-path parallel structure. This module effectively enhances small target details while suppressing background noise, with minimal increases in parameters and computational overhead.
4) Comprehensive experiments on three public datasets (DroneVehicle, low-light visible-infrared paired dataset (LLVIP), and FLIR infrared image super-resolution dataset (FLIR)) demonstrate that our cross-modal feature fusion and attention mechanism YOLO (CA-YOLO) model achieves superior mAP performance compared to existing single-modal and multimodal detection methods.

The remainder of this article is organized as follows. Section II provides a detailed overview of research related to multimodal fusion and small object detection, with a focus on analyzing the advantages and limitations of existing methods. Section III elaborates on the proposed algorithm for small object detection in drone remote sensing images, which is based on adaptive feature fusion and attention mechanisms. Section IV validates the effectiveness of the proposed method through experiments, comparing its performance against existing approaches, and conducting ablation studies to analyze the contributions of each module. Finally, Section V summarizes the entire work and discusses potential future research directions.

II. Related Work

A. Algorithms for Small Target Detection

As a significant challenge in the field of computer vision, the primary difficulties associated with small target detection can be categorized into three key aspects. First, small targets occupy a limited pixel area within images, resulting in insufficient feature information available for extraction. Second, these small targets are particularly vulnerable to interference from complex backgrounds, which further complicates the detection process. Finally, traditional detection algorithms often struggle to effectively extract robust features, thereby adversely affecting detection accuracy. With the rapid advancement of deep learning technologies, algorithms based on deep learning for small target detection have demonstrated considerable advantages in terms of both accuracy and robustness [15], [16].

In the field of deep learning, small target detection algorithms can be classified into two primary categories based on their reliance on the region proposal network (RPN): two-stage small target detection algorithms and one-stage small target detection algorithms. Two-stage methods employ an RPN to first generate candidate regions, followed by classification and localization within these regions. In contrast, single-stage methods adopt an end-to-end architecture that directly predicts bounding boxes and class labels without requiring a separate region proposal stage. This streamlined design offers superior computational efficiency, making it particularly suitable for real-time applications such as UAV systems [17].

The YOLO series has gained widespread adoption in UAV-based small target detection due to its high performance, precision, and architectural scalability. For example, Ni et al. [18] enhanced YOLOv8s by integrating a multiscale feature extraction module and optimizing the loss function, achieving improved small target detection performance at the cost of increased computational load. Li et al. [19] proposed SOD-YOLO, which refines the attention mechanism and incorporates multiscale feature fusion to simultaneously reduce model parameters and enhance accuracy; however, its generalization capability across diverse scenarios remains limited. Sun et al. [20] developed multi-YOLOv8 for IR moving target detection by fusing optical flow images and background-suppressed inputs with the current frame, thereby improving detection robustness—albeit with elevated computational complexity. Zhao et al. [21] introduced G-YOLO, a lightweight IR detection model based on YOLOv8, which optimizes the backbone network to reduce both parameter count and inference cost while improving small target detection accuracy. Nevertheless, its generalization performance requires further validation on broader datasets.

The small target detection algorithms for UAVs discussed above primarily focus on single-modal scenarios, specifically the detection of targets in either RGB or IR images. However, in practical applications, UAVs often face the necessity to conduct missions within complex and dynamic environments. In such situations, relying solely on single-modal image

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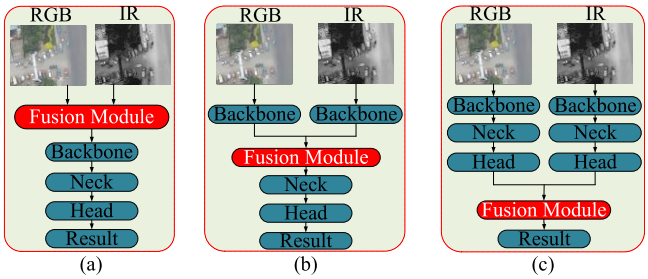


Fig. 1. Fusion methods. (a) Early stage, (b) mid-stage, and (c) late-stage fusions.

data may not provide sufficient information for accurate target detection. Notably, during nighttime or adverse weather conditions, the quality of RGB images significantly deteriorates, while IR images are susceptible to interference from thermal noise. As a result, researchers have begun exploring multimodal fusion techniques aimed at fully utilizing the complementary information present in both RGB and IR images to enhance the accuracy and robustness of target detection [22], [23]. Moreover, recent advancements in the YOLO series—specifically YOLOv10 [24], YOLOv11 [25], and YOLOv12 [26]—demonstrate superior performance compared to other variants of YOLO with respect to parameter count, computational complexity, and real-time processing capabilities. However, our experimental results indicate that YOLOv10 is particularly well suited for implementation within a dual-branch architecture tailored for UAV applications. Therefore, we propose an enhanced dual-branch small target detection algorithm for UAVs based on YOLOv10.

B. Multimodal Feature Fusion

Traditional multimodal feature fusion methods primarily encompass approaches based on multiscale transforms, sparse representations, subspace learning, and saliency-driven mechanisms [27]. However, these methods typically rely on handcrafted fusion rules, resulting in limited adaptability and poor generalization performance [28]. With the advancement of deep learning, end-to-end fusion frameworks have become dominant, enabling automatic extraction of intrinsic intermodal correlations and substantially reducing the need for manual intervention [29], [28]. The prevailing paradigm now adopts a three-stage fusion framework, comprising early, intermediate, and late fusion strategies [12], as illustrated in Fig. 1.

In the field of IR and RGB fusion, Wu et al. [30] pioneered the integration of the nonsubsampled shearlet transform with a deep Boltzmann machine. However, this method is hindered by high computational complexity and an excessive number of parameters. Subsequent studies have introduced several improvements, each accompanied by new tradeoffs. Qingyun et al. [31] proposed a dual-branch Transformer architecture based on self-attention [cross-modality fusion transformer (CFT)], capable of capturing global cross-modal dependencies, yet its overreliance on global context results in the degradation of local detail preservation. Dong et al. [32] presented FusionMamba, a state-space model-based approach that effectively mitigates modality discrepancies but suffers from a large parameter footprint and insufficient validation of generalization across diverse scenarios. Wu et al. [33] developed the multispectral interaction attention fusion model (MIAF), which incorporates multiscale feature extraction and detail enhancement mechanisms, yet its high computational cost limits its applicability in resource-constrained environments such as mobile deployment. Wang et al. [34] introduced the mask-guided Mamba fusion method (MGMF), demonstrating strong robustness under low-light conditions, but with a parameter count of 122 million, it compromises real-time inference capabilities.

In summary, existing methods typically face challenges related to the imbalance between global and local features, as well as the paradox of model complexity versus detection accuracy. In response to these issues, we propose an ACFF module. By optimizing the feature interaction mechanism, this module effectively retains the global modeling advantages of the Transformer while mitigating its computational burden. Furthermore, an adaptive local enhancement strategy is implemented to achieve a coordinated optimization of both detection accuracy and deployment efficiency.

C. Attention Mechanism

In the field of deep learning, attention mechanisms have gained widespread recognition for simulating the selective attention characteristic of biological vision systems and have become a key technique for enhancing feature extraction capabilities in models. Major types of attention include channel attention, spatial attention, self-attention, and hybrid attention [35], [36]. For instance, SENet [37] played a significant role in modeling interdependencies between channels. Saini et al. [38] introduced a mechanism specifically designed for spatial dimension refinement. CBAM [39] effectively integrated both channel and spatial focusing strategies. Furthermore, Transformer [40] successfully captured global contextual information through self-attention.

In the context of UAV-based small target detection, the application of attention mechanisms has led to notable progress in enhancing feature sensitivity; however, several technical bottlenecks remain. For example, Zeng et al. [41] proposed a spatial coordinate attention (SCA) module to improve the flow of feature information, yet its complex structure leads to a substantial increase in parameters and limited global feature modeling capability. Xiong et al. [42] developed a channel shuffling subspace attention module (CSSAM) with a lightweight design, but it suffers from insufficient channel representation and inadequate global information capture. In addition, Song et al. [43] introduced a multiscale hybrid attention (MHA) mechanism that enhances target features through a multibranch architecture, though deep stacking of such modules considerably increases computational complexity.

Current research reveals two major challenges in applying attention mechanisms to UAV small target detection: the conflict between complex model architectures and limited hardware resources, and the difficulty in balancing local feature enhancement with global information integration. To address these issues, this study proposes an LHA mechanism. By leveraging this approach, feature representation for small UAV targets can be effectively improved while reducing back-

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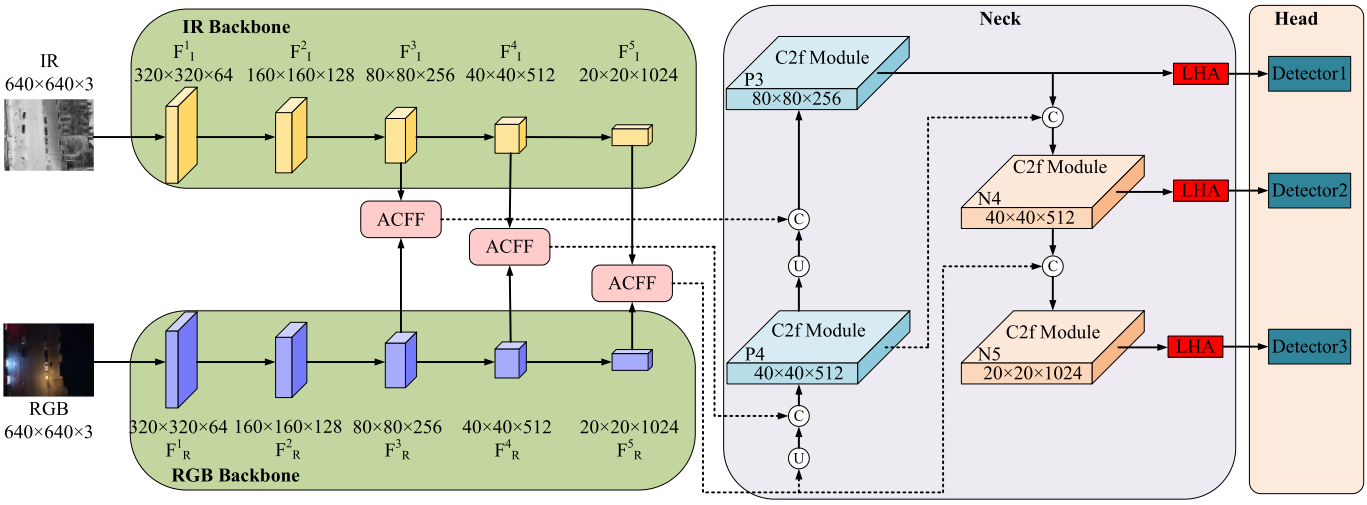


Fig. 2. Overall framework of CA-YOLO. The CA-YOLO consists of five main components. The first component is the input module, responsible for receiving RGB and IR images. The second component is the dual-branch backbone network, which employs a shared-parameter dual-branch architecture to independently extract hierarchical features from each modality. The third component is the ACFF module, which dynamically fuses the global semantic information and local detail information of RGB and IR modalities. The fourth component is the neck network, which further integrates multiscale features and combines LHA to further enhance the feature representation capability of small targets. The fifth component is the detection head, which generates the final detection outputs.

ground noise, maintaining relatively low computational costs, and minimizing the number of parameters.

III. Proposed Method

A. Overall Framework of CA-YOLO

To meet diverse task requirements, the YOLOv10 series offers six variants, ranging from YOLOv10n to YOLOv10x. These variants encompass a broad spectrum of parameter scales and detection accuracies, effectively achieving a balance between speed and precision. Considering the computational resource limitations inherent in UAV platforms, this article introduces an enhanced architecture based on YOLOv10n: the dual-branch multimodal small target detection network for UAVs (CA-YOLO). This architecture is specifically designed for RGB and IR multimodal small target detection tasks within UAV remote sensing imagery. As illustrated in Fig. 2, the overall framework of CA-YOLO is given as follows.

First, during the input stage, preprocessing is conducted on the collected RGB image I_{RGB} and IR image I_{IR} to ensure they meet the requirements for network input. Consequently, the preprocessed RGB image O_{RGB} and the IR image O_{IR} are obtained

$$I_{RGB} \rightarrow O_{RGB}, \quad I_{IR} \rightarrow O_{IR}. \tag{1}$$

In this context, I_{RGB}, O_{RGB}, I_{IR}, and $O_{IR} \in R^{W\times H\times C}$; and W, H, and C represent the width, height, and number of channels of the feature map, respectively.

Second, during the feature extraction phase, the preprocessed images O_{RGB} and O_{IR} are input into a dual-branch backbone network with shared parameters. This enables multilevel feature extraction. Specifically, it can be articulated as

$$F_R^{\mathrm{i}} = \psi_{backbone}^{RGB}\left(O_{RGB}; \theta_{RGB}^{i}\right); \quad F_I^{\mathrm{i}} = \psi_{backbone}^{IR}\left(O_{IR}; \theta_{IR}^{i}\right). \tag{2}$$

In this context, F_R^{i} and $F_I^{\mathrm{i}} \in R^{W\times H\times C}$ represent the feature maps of the ith layer ($i = 3, 4, 5$) within the RGB and IR branches, respectively. Meanwhile, θ_{RGB}^{i} and θ_{IR}^{i} denote the corresponding layers in these branches. In addition, $\psi_{backbone}^{RGB}(\cdot)$ and $\psi_{backbone}^{IR}(\cdot)$ refer to the backbone networks utilized for feature extraction from RGB images and IR images, respectively. Although RGB and IR modalities differ in imaging mechanisms, they share fundamental structural characteristics crucial for object detection, such as object boundaries, shapes, and motion patterns. The parameter-sharing strategy focuses the backbone on extracting these common low-level geometric features, acting as a regularization mechanism to prevent overfitting to a single-modality. Furthermore, this design reduces the backbone parameter count by approximately half compared to pseudo-siamese networks, which is critical for resource-constrained UAV platforms. The modality-specific discrepancies are subsequently handled by the ACFF module, which dynamically weights independent features after the backbone extraction.

Third, during the multimodal feature fusion stage, the extracted feature pairs F_R^{i} and F_I^{i} are input into the *ACFF* module. Through a dynamic adjustment mechanism, both global semantic information and local detailed features from RGB and IR modalities are integrated, resulting in the fused feature F_{Fused}^{i}. The definition of this fused feature is provided by the following equation:

$$F_{Fused}^{i} = ACFF\left(F_{RGB}^{i}, F_{IR}^{i}\right) \tag{3}$$

where F_{Fused}^{i} denotes the fused features.

Fourth, during the neck network stage, the fused feature F_{Fused}^{i} undergoes multiscale feature integration and enhancement facilitated by the neck network. This process further optimizes the hierarchical structure of the feature pyramid. In addition, an LHA mechanism is incorporated into the enhanced and integrated features to enhance detection capabilities for small targets across various scales. Consequently, an optimized feature representation $F_{Enhanced}^{i}$ is derived, which can be expressed as

$$F_{Enhanced}^{i} = LHA\left(\xi_{Neck}\left(F_{Fused}^{i}\right)\right) \tag{4}$$

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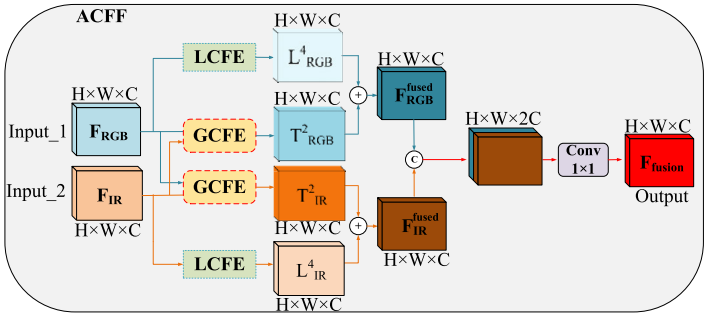


Fig. 3. ACFF module. ACFF consists of two parallel submodules: the GCFE branch and the LCFE branch. The GCFE branch captures long-range dependencies by leveraging a cross-modal attention mechanism, thereby enhancing the representation of global semantic information. The LCFE branch employs cosine similarity to measure local feature correlations, thereby enhancing the contrast between foreground objects and background regions.

where $F^{i}_{Enhanced} \in R^{W\times H\times C}$ denotes the optimized features, while $\xi_{Neck}(\cdot)$ refers to the neck network.

Finally, during the detection phase, the optimized feature $F^{i}_{Enhanced}$ is input into the detection head module. Through the coordinated functioning of the convolutional layer, regression layer, and classification layer, the final detection result $R_{Detector}$ is generated. Its definition is provided by the following equation:

$$R_{Detector} = \phi_{Head}\left(F^{i}_{Fenhanced}\right) \tag{5}$$

where $R_{Detector}$ denotes the detection outcome and $\phi_{Head}(\cdot)$ signifies the detection head module within the detection network.

B. Adaptive Cross-Modal Feature Fusion

In the field of UAV remote sensing, traditional single-modal small target detection methods often suffer from missed detections and false alarms in complex scenarios. In contrast, multimodal approaches can effectively mitigate these issues by leveraging complementary information. Shen et al. [44] introduced the cross-modal feature enhancement (CFE) module, which captures and fuses features from various modalities from a global perspective. However, this approach overlooks the importance of capturing local features, which are particularly critical for detecting small targets. To address this limitation, this article proposes the ACFF module that integrates local and global features in a collaborative manner, as illustrated in Fig. 3. The detailed process is outlined as follows.

First, the feature maps $F_{RGB} \in R^{H\times W\times C}$ and $F_{IR} \in R^{H\times W\times C}$, which are input by the RGB modality branch and the IR modality branch, respectively, yield enhanced feature maps $T^{2}_{RGB} \in R^{H\times W\times C}$ and $T^{2}_{IR} \in R^{H\times W\times C}$ after being processed through the global CFE (GCFE) module. Specifically, it can be expressed as

$$T^{2}_{RGB}, T^{2}_{IR} = GCFE\left(F_{RGB}\right), \quad GCFE\left(F_{IR}\right). \tag{6}$$

Second, the feature maps $F_{RGB} \in R^{H\times W\times C}$ and $F_{IR} \in R^{H\times W\times C}$, which are derived from the RGB modality branch and the IR modality branch, respectively, undergo processing through the local cosine similarity feature enhancement (LCFE) module. This results in the generation of enhanced feature maps $L^{4}_{RGB} \in R^{H\times W\times C}$ and $L^{4}_{IR} \in R^{H\times W\times C}$. Specifically, it can be expressed as

$$L^{4}_{RGB} = LCFE\left(F_{RGB}\right), \quad L^{4}_{IR} = LCFE\left(F_{IR}\right). \tag{7}$$

Third, to further integrate global and local information, the enhanced feature maps generated by the GCFE module and the LCFE module undergo an addition and fusion operation across different modal branches. Consequently, the feature maps $F^{fused}_{RGB} \in R^{H\times W\times C}$ and $F^{fused}_{IR} \in R^{H\times W\times C}$, resulting from the fusion of the RGB modality branch and the IR modality branch, are obtained, respectively,

$$\begin{cases} F^{fused}_{RGB} = T^{2}_{RGB} + L^{4}_{RGB} \\ F^{fused}_{IR} = T^{2}_{IR} + L^{4}_{IR}. \end{cases} \tag{8}$$

Finally, the feature maps F^{fused}_{RGB} and F^{fused}_{IR} obtained are concatenated along the channel dimension. Subsequently, this concatenated feature map is processed through a convolutional layer with a kernel size of one to achieve channel dimensionality reduction, resulting in the final feature map F_{fusion}

$$F_{fusion} = Conv_{1\times 1}\left(Cat\left(F^{fused}_{RGB}, F^{fused}_{IR}\right)\right). \tag{9}$$

1) Global CFE: The core objective of the GCFE module is to establish global feature associations between RGB and IR modalities while minimizing the computational overhead associated with purely global modeling. Taking the RGB branch as an example, as shown in Fig. 4, the input RGB and IR feature maps $F_{RGB} \in R^{H\times W\times C}$ and $F_{IR} \in R^{H\times W\times C}$ are initially processed using a modified hybrid pooling strategy to minimize detail loss (see Appendix A). Adaptive fusion is achieved by introducing a learnable parameter α, which generates compressed channel feature vectors $T_{RGB} \in R^{HW\times C}$ and $T_{IR} \in R^{HW\times C}$. Subsequently, key and value matrices for T_{RGB}, along with a query matrix for T_{IR}, are generated through linear projection operations. Following similarity calculations and softmax normalization, a cross-modal attention weight matrix Z_R is derived. This matrix Z_R is then fused with T_{RGB} through a weighted combination to obtain the feature representation T'_{RGB}. The representation $T^{1}_{RGB} \in R^{HW\times C}$ is further enhanced by employing a feedforward network (FFN), ultimately yielding the globally enhanced feature $T^{2}_{RGB} \in R^{H\times W\times C}$ that is suitable for small object detection tasks. For

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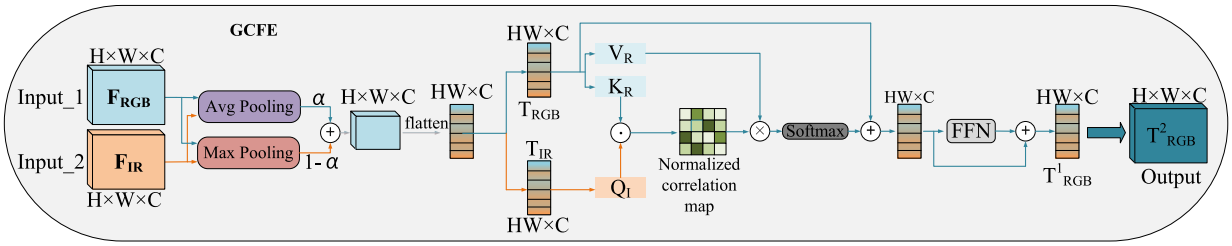


Fig. 4. GCFE module structure diagram. GCFE core lies in capturing long-range dependencies through cross-modal self-attention.

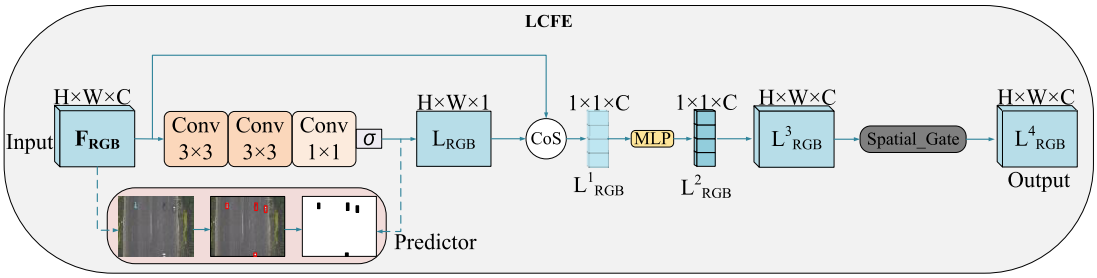


Fig. 5. LCFE module structure diagram. LCFE is designed to strengthen the representation of local features for small targets by integrating three key components: box-level masking, cosine similarity, and spatial gating.

detailed mathematical derivations and formula explanations pertaining to this module, please refer to Appendix A.

2) Local Cosine Similarity Feature Enhancement: The LCFE module is specifically designed to enhance the feature representation of local target regions and improve the discriminability between small targets and complex backgrounds. As shown in Fig. 5, using the RGB branch as an example, the process commences with the generation of a box-level mask $L_{RGB} \in R^{H\times W\times 1}$ by applying a convolutional operation to the feature map F_{RGB} of the RGB branch, followed by a sigmoid activation function based on annotated bounding box information. Subsequently, cosine similarity between the mask $L_{RGB} \in R^{H\times W\times 1}$ and feature map F_{RGB} is computed to derive the correlation matrix $L^1_{RGB} \in R^{1\times 1\times C}$. This matrix is then processed through a multilayer perceptron (MLP) for a nonlinear transformation that enhances its representational capacity, resulting in a refined correlation matrix $L^2_{RGB} \in R^{1\times 1\times C}$. Matrix L^2_{RGB} is subsequently element-wise multiplied with the original feature map F_{RGB} to produce an intermediate enhanced feature matrix $L^3_{RGB} \in R^{H\times W\times C}$. Finally, a spatial gating module is employed to further refine local spatial attention within the feature matrix L^3_{RGB}, culminating in the final enhanced feature map $L^4_{RGB} \in R^{H\times W\times C}$. For detailed mathematical derivations and formula explanations pertaining to this module, please refer to Appendix B.

C. Lightweight Hybrid Attention

To address the prevailing challenges of background noise interference and target feature dilution in small target detection within UAV remote sensing images, this study innovatively proposes an LHA mechanism incorporating a channel rearrangement strategy. As illustrated in Fig. 6, this mechanism is structurally optimized based on the ACMix framework [45]. By establishing a global–local attention architecture via dual-branch collaborative optimization, it effectively addresses the critical challenge whereby small target features are prone to being overwhelmed by complex surface backgrounds. Specifically, the detailed operations are given as follows:

$$F_{out} = Shuffle\left(F^1 + F^2\right), \quad F_{out} \in R^{H\times W\times C} \tag{10}$$

where F_{out} denotes the output feature map; *Shuffle*$(\cdot)$ denotes the channel shuffle function; and F^1 and F^2 denote the feature maps generated after global attention and local attention, respectively.

1) Global Branch: Finally, for an input feature map $F \in R^{H\times W\times C}$, a projection is performed through three parallel 1×1 convolution operations, resulting in the projection matrices *query*, *key*, and *value*. Subsequently, 7×7 sliding window operations are applied to the obtained projection matrices *key* and *value*, producing the window matrices K^1 and V^1, respectively,

$$\begin{cases} Query = Conv_{1\times 1}(F), & Query \in R^{H\times W\times (C/4)} \\ Key = Conv_{1\times 1}(F), & Key \in R^{H\times W\times (C/4)} \\ Value = Conv_{1\times 1}(F), & Value \in R^{H\times W\times (C/4)} \end{cases} \tag{11}$$

$$\begin{cases} K^1 = Unfold_{7\times 7}(Key), & K^1 \in R^{H\times W\times 7^2\times (C/4)} \\ V^1 = Unfold_{7\times 7}(Value), & V^1 \in R^{H\times W\times 7^2\times (C/4)}. \end{cases} \tag{12}$$

Second, the position function and a 1×1 convolution operation are applied to the *query* projection matrix in order to extract the position encoding information *pe* of the *query*. Subsequently, an unfold operation is performed on *pe* for expansion, resulting in the relative position encoding *rpe*, which enables coverage of all domain positions

$$\begin{cases} pos = position(Query), & pos \in R^{2\times H\times W} \\ pe = Conv_{1\times 1}(pos), & pe \in R^{(C/4)\times H\times W} \\ rpe = Unfold_{7\times 7}(pe), & rpe \in R^{(C/4)\times 7^2\times H\times W}. \end{cases} \tag{13}$$

Finally, the positional encoding *pe* is superimposed onto the window matrix K^1, followed by subtracting the relative

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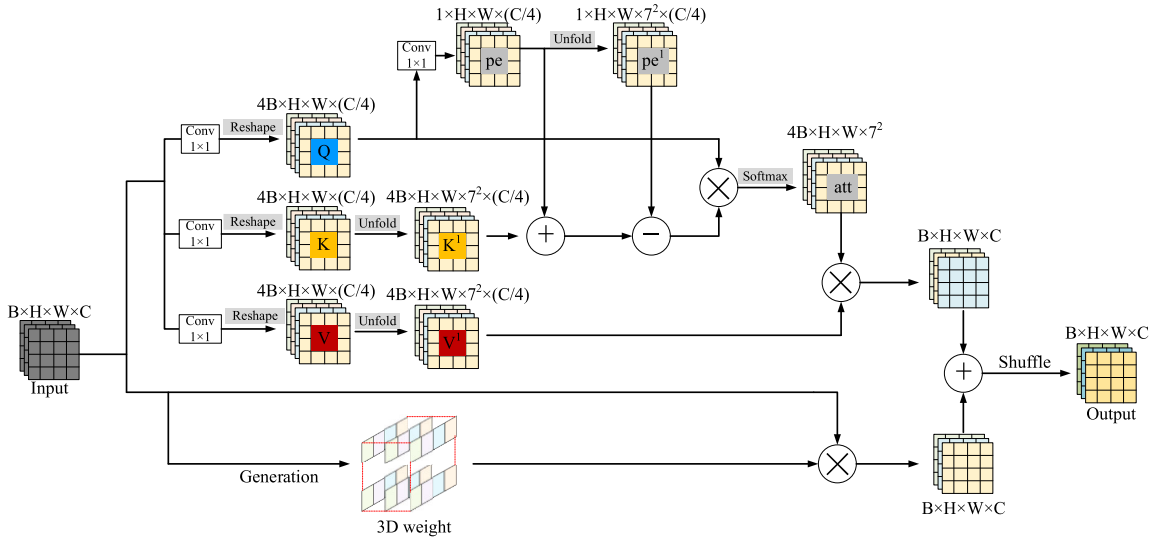


Fig. 6. LHA module structure diagram. The LHA module consists of two parallel branches: a global branch and a local branch. The global branch utilizes a sliding window-based self-attention mechanism to effectively capture global contextual information. The local branch employs a variance-based attention mechanism to enhance the feature responses of prominent local regions. The fused output from the two branches is further refined through the channel shuffle module, enabling comprehensive integration of complementary global and local features across the channel dimension.

positional encoding *rpe*, thereby generating a window matrix embedded with relative positional encoding. Thereafter, K^2 is multiplied by the *query* projection matrix, and the resulting similarity matrix is normalized via the *soft*max function to derive the attention scores *att*. Concurrently, the window matrix V^1 is weighted by the obtained attention scores *att*, yielding the output F^1 of the global attention branch

$$\begin{cases} att = Soft\max\left(\dfrac{Query \times \left(pe + K^1 - rpe\right)}{\sqrt{D}}\right), & att \in R^{H\times W\times 7^2} \\ F^1 = att \times V^1, & F^1 \in R^{H\times W\times C} \end{cases} \tag{14}$$

where D denotes the feature dimension of *query*.

2) Local Branch: First, compute the spatial mean μ for each channel in the input feature map $F(i, j)$. In addition, calculate the squared difference $D(i, j)$ between each position and its corresponding mean

$$\mu = \frac{1}{H \times W}\sum_{i=1}^{H}\sum_{j}^{W} F(i, j), \quad \mu \in R^{1\times 1\times C} \tag{15}$$

$$D(i, j) = (F(i, j) - \mu)^2, \quad D(i, j) \in R^{H\times W\times C} \tag{16}$$

where H and W denote the height and width of the spatial dimension of the feature map, respectively.

Second, we compute the sum of $D(i, j)$ across the spatial dimension and normalize it by the degrees of freedom n to obtain an unbiased estimator denoted as *Var*. Subsequently, based on the mathematical relationship between *Var* and $D(i, j)$, we derive the attention weight δ

$$Var = \frac{1}{n}\sum_{i=1}^{H}\sum_{j=1}^{W} D(i, j), \quad Var \in R^{1\times 1\times C} \tag{17}$$

$$\delta = \sigma\left(\frac{D(i, j)}{4\,(Var + \lambda)} + 0.5\right) \tag{18}$$

where $n = H + W - 1$, λ denotes the smoothing coefficient, and σ denotes the sigmoid activation function.

Finally, multiply the attention weights by the original feature map on an element-by-element basis to enhance the features of significant regions

$$F^2 = F \times \delta, \quad F^2 \in R^{H\times W\times C}. \tag{19}$$

IV. Experiments

In this section, we perform a systematic and comprehensive validation of the proposed method across five key aspects: dataset and evaluation metrics, implementation details, comparative experiments, ablation studies, and visualization analysis. The objective is to ensure the effectiveness, stability, and superiority of the proposed approach.

A. Datasets and Evaluation Metrics

The DroneVehicle dataset [46], released by Tianjin University in 2020, is a comprehensive multimodal benchmark for UAV detection. It comprises 28 439 pairs of aligned RGB and IR images, capturing both daytime and nighttime scenarios to enhance environmental adaptability. The annotations within the dataset include a total of 441 642 vehicle instances across three target categories: cars, trucks, and buses. Specifically, the training set consists of 17 990 pairs of aligned RGB and IR images; the validation set contains 1469 pairs; and the test set includes 8980 pairs of aligned RGB and IR images.

The LLVIP dataset [47] is a multispectral pedestrian detection dataset that comprises paired visible and IR images specifically designed for low-light vision applications. Released by Beijing University of Posts and Telecommunications in 2021, the dataset includes a total of 15 488 pairs of RGB and IR images, amounting to 30 976 individual

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TABLE I
EXPERIMENTAL CONFIGURATION

Parameters	Value
Epochs	100
Batch_size	4
Input image size	640×640
Gradient Optimizer	SGD
Momentum	0.935
Initial learning rate	0.01
Final learning rate	0.00001
Data enhancement	Mosaic
IoU	0.5

images. Notably, the training set consists of 12 025 pairs of aligned RGB and IR images, while the validation set contains 3463 pairs of aligned RGB and IR images.

The FLIR dataset [48] was utilized in this study. To ensure consistency in the scale of the two modalities of images within the FLIR dataset, we conducted our experiments using the "aligned" version. The "aligned" FLIR dataset comprises 5142 pairs of RGB and IR images, with 4129 pairs designated for training and 1013 pairs allocated for testing. This dataset encompasses a variety of urban street scenes and includes three object categories: bicycles, cars, and people.

For performance evaluation, average precision (AP) is adopted as the primary metric. AP provides a comprehensive assessment of detection performance by computing the area under the precision–recall (P–R) curve through interpolation at various recall levels. Positive detections are typically determined using an intersection over union (IoU) threshold, commonly set at 0.5. The mean of AP values across all object categories, denoted as mAP, is used to evaluate the overall detection accuracy. In addition, we utilize frames per second (FPS) to evaluate the inference speed, calculated as $1/t_{lat}$, where t_{lat} includes preprocessing, inference, and NMS time. To comprehensively assess detection reliability, we employ miss rate (MR) and false alarm rate (FAR). MR is defined as the ratio of missed targets to the total number of ground truth objects, and FAR represents the ratio of false detections to the total number of predicted positive samples. They are calculated as follows:

$$MR = \frac{FN}{TP + FN} \tag{20}$$

$$FAR = \frac{FP}{TP + FP} \tag{21}$$

where FN, TP, and FP denote the false negatives, true positives, and false positives, respectively.

B. Implementation Details

To ensure the reliability and stability of the proposed algorithm, consistent parameter settings were employed across all comparative and ablation experiments. The experimental environment for our method was configured on the Windows 10 operating system. The hardware specifications include an Intel i7-14700KF processor, 16 GB of RAM, and an NVIDIA RTX 4080 SUPER GPU with 16 GB of video memory. At the software level, CUDA 11.8 was utilized in conjunction with the PyTorch framework (v2.0.0) and Python 3.10. Notably, no pretrained weights were used during the model training phase. A detailed overview of the key parameter settings for model training is provided in Table I.

C. Experiments With Different Algorithms

In order to thoroughly validate the superiority of the proposed CA-YOLO algorithm, we conducted a comprehensive analytical comparison of 12 mainstream methods currently prevalent in the field. Among these, eight are single-modality detection algorithms [24], [25], [26], [49], [50], [51], [52], [53]; the remaining six are multimodality detection algorithms [44], [46], [54], [55], [56], [57]. Furthermore, under consistent experimental conditions and evaluation metrics, we assessed the performance of each method. Tables II–IV present the experimental results obtained from the DroneVehicle dataset, LLVIP dataset, and FLIR dataset, respectively.

The quantitative analysis results presented in Table II demonstrate that the proposed CA-YOLO algorithm achieves state-of-the-art detection performance on the DroneVehicle dataset, attaining an mAP of 75.1%, which significantly surpasses both existing single-modal and multimodal approaches. In terms of computational efficiency, CA-YOLO operates with only 6.8 MB of model parameters and 16.4 giga floating-point operations per second (GFLOPs) of computational cost, representing a 93.3% reduction in parameter count and an 81.8% decrease in computational complexity relative to C^2former, the current leading multimodal method. With an inference latency of 17.5 ms (equivalent to 57.1 frames/s), CA-YOLO exhibits strong real-time inference capabilities.

1) In the RGB modality, CA-YOLO improves mAP by 6.2 percentage points over the baseline YOLOv10n (68.9%) and by 4.8 percentage points compared to the top-performing single-modal model YOLOv11n (70.3%).
2) In the IR modality, CA-YOLO improves mAP by 7.3 percentage points compared to YOLOv10n (67.8%) and by 5.6 percentage points relative to the top-performing single-modal IR detector YOLOv8n (69.5%).
3) Compared to the state-of-the-art multimodal method C^2former, which reports an mAP of 72.6%, CA-YOLO achieves a 2.5 percentage point improvement while demonstrating superior performance in model compactness, computational efficiency, and inference speed. Notably, the multimodal method UA-CMDet achieves an mAP of only 64.0% in this evaluation, underperforming several YOLO-based single-modal models, despite having a substantially larger parameter footprint of 144.5 MB and a lower inference speed of 53.8 frames/s. This result highlights the inherent challenges in multimodal data fusion, indicating that the practical effectiveness of such approaches is highly dependent on both the specific application scenario and the design sophistication of the fusion strategy.

The quantitative analysis results on the LLVIP dataset, as presented in Table III, demonstrate that CA-YOLO achieves an AP of 96.1%, delivering state-of-the-art detection performance and significantly outperforming existing single-modal and

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TABLE II
COMPARATIVE ANALYSES OF DIFFERENT ALGORITHMS ON THE DRONEVEHICLE DATASET

Method	Modality		Car	freight Car	truck	bus	van	Params. (MB)	GFLOPs	Inference (ms)	FPS	mAP(%)
	RGB	IR										
RetinaNet[49]	✓	×	66.4	12.6	27.1	61.1	18.2	32.5	144.3	53.5	18.7	37.1
Faster RCNN[50]	✓	×	66.8	25.2	37.4	65.3	22.1	41.4	157.6	39.2	25.5	43.4
SSD[51]	✓	×	64.0	10.5	25.8	58.3	16.0	24.3	254.3	27.0	37.0	34.9
YOLOv5n[52]	✓	×	91.4	48.2	60.8	92.3	53.3	2.4	7.1	5.0	200.0	69.2
YOLOv8n[53]	✓	×	92.0	48.4	62.2	92.0	53.8	2.9	8.1	1.8	555.6	69.7
YOLOv10n[24]	✓	×	91.5	50.2	59.9	91.3	51.7	2.6	8.2	2.7	370.4	68.9
YOLOv11n[25]	✓	×	92.0	49.8	62.9	91.9	54.9	2.5	6.3	2.5	400.0	70.3
YOLOv12n[26]	✓	×	91.6	51.0	61.9	92.7	49.5	2.4	5.8	3.3	303.0	69.3
RetinaNet[49]	×	✓	78.4	26.2	31.6	66.5	15.3	32.5	144.3	12.5	79.8	43.6
Faster RCNN[50]	×	✓	87.7	33.6	41.2	76.5	27.2	41.4	157.6	25.5	39.2	53.2
SSD[51]	×	✓	70.5	18.3	29.7	63.8	14.0	26.2	254.3	55.1	18.1	39.3
YOLOv5n[52]	×	✓	92.3	47.5	54.9	93.5	49.2	2.4	7.1	5.4	185.2	67.5
YOLOv8n[53]	×	✓	92.4	48.1	61.6	93.3	48.1	2.9	8.1	2.2	454.5	69.5
YOLOv10n[24]	×	✓	92.3	47.3	59.9	92.2	47.6	2.6	8.2	2.7	370.4	67.8
YOLOv11n[25]	×	✓	92.3	47.2	59.4	93.5	52.1	2.5	6.3	2.6	384.6	68.9
YOLOv12n[26]	×	✓	92.2	48.7	52.2	93.6	49.4	2.4	5.8	3.2	312.5	67.2
TFDet[54]	✓	✓	93.6	47.3	62.3	92.9	47.4	50.7	135.2	6.5	153.8	68.7
UA-CMDet[46]	✓	✓	87.5	46.8	60.7	87.1	38.0	144.5	234.7	18.6	53.8	64.0
C^2former[55]	✓	✓	92.7	53.8	69.1	90.7	56.5	100.8	89.9	15.6	64.1	72.6
DACFusion[56]	✓	✓	92.3	56.1	68.7	90.3	53.4	32.9	46.8	10.3	97.1	72.2
ICAFusion[44]	✓	✓	93.1	54.4	69.6	90.4	55.1	120.3	192.6	12.9	77.5	72.5
CFFDNet[57]	✓	✓	92.6	55.9	68.6	90.6	53.8	316.4	411.7	57.1	17.5	72.3
CA-YOLO (Our)	✓	✓	97.1	57.1	68.2	95.2	58.0	6.8	16.4	17.5	57.1	75.1

TABLE III
COMPARATIVE ANALYSES OF DIFFERENT ALGORITHMS ON THE LLVIP DATASET

Method	Modality		Params (M)	GFLOPs	Inference (ms)	FPS	AP(%)
	RGB	IR					
RetinaNet[49]	✓	×	32.5	144.3	17.8	56.2	83.2
Faster RCNN[50]	✓	×	41.4	157.6	9.9	100.7	86.7
SSD[51]	✓	×	26.2	72.4	9.6	104.2	78.3
YOLOv5n[52]	✓	×	2.4	7.1	5.6	178.6	88.2
YOLOv8n[53]	✓	×	2.9	8.1	3.2	312.5	89.4
YOLOv10n[24]	✓	×	2.6	8.2	3.6	277.8	87.3
YOLOv11n[25]	✓	×	2.5	6.3	3.3	303.0	88.2
YOLOv12n[26]	✓	×	2.4	5.8	4.0	250.0	89.4
RetinaNet[49]	×	✓	32.5	144.3	10.2	98.0	88.1
Faster RCNN[50]	×	✓	41.4	157.6	9.2	108.7	91.4
SSD[51]	×	✓	26.2	72.4	10.6	94.3	84.6
YOLOv5n[52]	×	✓	2.4	7.1	5.6	178.6	94.6
YOLOv8n[53]	×	✓	2.9	8.1	3.1	322.6	94.2
YOLOv10n[24]	×	✓	2.6	8.2	3.4	294.1	94.4
YOLOv11n[25]	×	✓	2.5	6.3	3.1	322.6	94.4
YOLOv12n[26]	×	✓	2.4	5.8	5.1	196.1	94.6
TFDet[54]	✓	✓	50.7	135.2	6.3	158.7	95.2
UA-CMDet[46]	✓	✓	144.5	234.7	11.6	86.2	95.3
C^2former[55]	✓	✓	100.8	89.9	12.1	82.6	95.6
DACFusion[56]	✓	✓	32.9	46.8	13.3	75.2	94.2
ICAFusion[44]	✓	✓	120.3	192.6	14.6	68.5	95.2
CFFDNet[57]	✓	✓	316.4	411.7	21.4	46.7	95.6
CA-YOLO (Our)	✓	✓	6.8	16.4	17.2	58.1	96.1

multimodal detection methods. In terms of model efficiency, CA-YOLO exhibits substantially fewer parameters and lower computational complexity compared to other multimodal approaches, resulting in a significantly reduced computational burden. With respect to inference speed, CA-YOLO requires 17.2 ms per inference, achieving a frame rate of 58.1 frames/s. Although, as a multimodal method, its inference speed is slower than that of the single-modal YOLO series models,

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TABLE IV
COMPARATIVE ANALYSES OF DIFFERENT ALGORITHMS ON THE FLIR DATASET

Method	Modality		bicycle	person	car	Params (M)	GFLOPs	Inference (ms)	FPS	mAP(%)
	RGB	IR								
RetinaNet[49]	✓	×	37.5	50.0	67.2	32.5	144.3	5.3	188.7	51.6
Faster RCNN[50]	✓	×	41.6	52.9	69.6	41.4	157.6	5.3	188.7	54.7
SSD[51]	✓	×	34.2	45.8	65.4	26.2	72.4	12.3	81.3	48.5
YOLOv5n[52]	✓	×	34.1	55.1	77.2	2.4	7.1	2.2	454.5	55.5
YOLOv8n[53]	✓	×	40.3	59.0	77.3	2.9	8.1	1.1	909.1	58.9
YOLOv10n[24]	✓	×	33.7	53.0	74.6	2.6	8.2	1.5	666.7	53.8
YOLOv11n[25]	✓	×	42.1	60.3	78.1	2.5	6.3	1.4	714.3	60.1
YOLOv12n[26]	✓	×	35.4	57.1	76.7	2.4	5.8	1.8	555.6	56.4
RetinaNet[49]	×	✓	47.1	73.3	79.2	32.5	144.3	5.2	192.3	66.5
Faster RCNN[50]	×	✓	50.4	75.7	82.6	41.4	157.6	5.2	192.3	69.6
SSD[51]	×	✓	43.8	68.5	76.3	26.2	72.4	5.2	192.3	62.9
YOLOv5n[52]	×	✓	54.8	74.0	84.7	2.4	7.1	2.2	454.5	70.2
YOLOv8n[53]	×	✓	50.3	76.8	88.3	2.9	8.1	1.1	909.1	70.8
YOLOv10n[24]	×	✓	50.9	75.4	85.9	2.6	8.2	1.7	588.2	70.7
YOLOv11n[25]	×	✓	52.1	77.6	87.2	2.5	6.3	1.2	833.3	70.3
YOLOv12n[26]	×	✓	48.2	77.7	87.0	2.4	5.8	2.5	400.0	70.0
TFDet[54]	✓	✓	49.8	76.0	87.1	50.7	135.2	7.6	131.6	71.0
UA-CMDet[46]	✓	✓	56.5	75.4	80.6	144.5	234.7	17.4	57.5	70.8
C^{2}former[55]	✓	✓	52.4	76.7	84.3	100.8	89.9	21.5	46.5	71.1
DACFusion[56]	✓	✓	55.3	74.6	82.6	32.9	46.8	12.0	83.3	70.8
ICAFusion[44]	✓	✓	53.5	75.8	84.2	120.3	192.6	35.0	28.6	71.2
CFFDNet[57]	✓	✓	54.8	75.9	83.1	316.4	411.7	22.3	44.8	71.3
CA-YOLO (Our)	✓	✓	53.8	77.1	83.3	6.8	16.4	17.7	56.5	71.4

it still surpasses mainstream multimodal competitors such as ICAFusion and CFFDNet.

1) In the RGB modality, CA-YOLO's AP exceeds that of the baseline model YOLOv10n (87.3%) by 8.8 percentage points and outperforms the current best single-modal model YOLOv12n (89.4%) by 6.7 percentage points.
2) In the IR modality, CA-YOLO improves AP by 1.7 percentage points over YOLOv10n (94.4%) and by 1.5 percentage points relative to YOLOv12n (94.6%), the top-performing single-modal model.
3) Compared with the current state-of-the-art multimodal method C^{2}former (AP 95.6%), CA-YOLO achieves a 0.5 percentage point gain in AP, while its parameter count amounts to only 6.7% of C^{2}former's, and its computational complexity is reduced to 16.4 GFLOPs. Despite a slightly lower inference speed than C^{2}former, CA-YOLO sets a new benchmark among multimodal detection methods by achieving superior performance in detection accuracy, model compactness, and computational efficiency.

As shown in the quantitative analysis results on the FLIR dataset presented in Table IV, CA-YOLO maintains state-of-the-art detection performance with an mAP of 71.4% while demonstrating notable advantages in model complexity and inference efficiency. The detailed analysis is given as follows.

1) In the RGB modality, CA-YOLO achieves an mAP that is 17.6 percentage points higher than that of the baseline model YOLOv10 (53.8%) and 11.3 percentage points higher than the current best single-modal model YOLOv11n (60.1%).
2) In the IR modality, CA-YOLO outperforms the baseline model YOLOv10n (70.7%) by 0.7 percentage points and surpasses the top-performing single-modal model YOLOv8n (70.8%) by 0.6 percentage points.
3) Compared to the current state-of-the-art multimodal method C^{2}former, CA-YOLO achieves a 0.3 percentage point improvement in mAP while exhibiting significant reductions in parameter count and computational complexity relative to other multimodal approaches. With an inference time of 17.7 ms and a frame rate of 56.7 frames/s, CA-YOLO improves upon C^{2}former's speed by 17.7%. Although its inference speed remains slower than some multimodal methods, CA-YOLO delivers the best overall performance when considering detection accuracy, model compactness, and computational efficiency in a balanced manner. It should be noted that in this study, the TFDet algorithm from [54] achieves an mAP of 71.0%, which is lower than the value reported in the original paper. This disparity stems from the fact that the original method employs SwinTransformer-Tiny as the backbone network on the FLIR dataset. However, considering the availability of the official code and aiming to avoid potential biases introduced by self-implementation, we decided to adopt the backbone network configuration CSPDarknet53 provided by the

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TABLE V
ANALYSIS ON THE VALIDITY OF THE DUAL-BRANCH YOLOV 10N ARCHITECTURE

Baseline	Params. (MB)	GFLOPs	mAP(%)
Double-branch YOLOv10n [24]	3.91	11.6	71.8
Double-branch YOLOv11n [25]	3.94	11.4	70.4
Double-branch YOLOv12n [26]	4.23	11.3	68.1
Double-branch SwinTransformer[58]	5.19	15.2	57.2
Double-branch ConvNeXt [59]	2.88	9.1	51.9
Double-branch FasterNet [60]	1.76	6.9	62.9

official code. This decision has influenced the deviation in the TFDet results and also highlights the significant impact of backbone network selection on object detection performance.

D. Ablation Experiment

In order to comprehensively assess the specific contributions of each key component within the proposed CA-YOLO algorithm to the model's performance, this section designs and carries out a series of ablation experiments using the DroneVehicle dataset. Subsequently, an in-depth analysis of the experimental results is conducted.

1) Impact of Different Dual-Branch Architectures on the Model Performance: To validate the performance superiority of the dual-branch YOLOv10 [24] under the shared-parameter mechanism, this study conducts comparative evaluations with structurally similar models, including YOLOv11 [25], YOLOv12 [26], SwinTransformer [58], ConvNeXt [59], and FasterNet [60]. To ensure the reliability and comparability of the experimental results, all models employ the add feature fusion method consistently. As presented in Table V, the parameter count and computational complexity of the dual-branch YOLOv10n are 3.91 MB and 11.6 GFLOPs, respectively; for dual-branch YOLOv11n, these values are 3.94 MB and 11.4 GFLOPs; and for dual-branch YOLOv12n, they are 4.23 MB and 11.3 GFLOPs. Among the evaluated models, the dual-branch SwinTransformer exhibits the highest parameter count (5.19 MB) and computational complexity (15.2 GFLOPs), while the dual-branch ConvNeXt has the lowest parameter count (2.88 MB) and moderate computational complexity (9.1 GFLOPs). The dual-branch FasterNet demonstrates the minimum in both metrics, with 1.76 MB of model parameters and 6.9 GFLOPs of computational complexity. Although the parameter counts and computational costs of dual-branch YOLOv10n, YOLOv11n, and YOLOv12n are comparable, their object detection performance, measured by mAP, shows notable differences: dual-branch YOLOv10n achieves an mAP of 71.8%, surpassing dual-branch YOLOv11n (70.4%) and dual-branch YOLOv12n (68.1%). In comparison, the mAP values for dual-branch SwinTransformer, ConvNeXt, and FasterNet are 57.2%, 51.9%, and 62.9%, respectively. These experimental findings indicate that the dual-branch architecture of YOLOv10 delivers superior object detection performance with relatively low computational overhead, highlighting its potential as an efficient foundational architecture for model design.

TABLE VI
ANALYSIS OF THE EFFECTIVENESS OF THE ASFF FUSION MODULE

Method		Params. (MB)	GFLOPs	mAP (%)	MR (%)	FAR (%)
+CFT[31]		12.2	13.7	53.2	30.5	4.5
+DMFF[44]		6.2	13.0	68.1	29.8	4.2
+ CMDAF [61]		3.9	11.6	70.7	29.1	3.9
+FFM [54]		4.4	12.3	70.4	30.2	4.4
+ Fusion[62]		4.0	11.9	69.7	30.0	4.3
+ACFF	+GCFE	5.7	13.5	71.9	28.1	3.5
	++LCFE (ACFF)	6.5	15.9	72.3	26.8	2.8

2) Impact of the ACFF Fusion Module on the Model Performance: To systematically evaluate the performance advantages of the ACFF module in UAV small object detection, we conducted multidimensional experiments based on the dual-branch YOLOv10 baseline architecture. As shown in Table VI, comparative analyses were performed among five feature fusion strategies (CFT [31], dual-modal feature fusion (DMFF) [43], cross-modality differential aware fusion (CMDAF) [61], feature fusion module (FFM) [54], and FusionModule [62]) alongside the proposed ACFF module (i.e., GCFE fused with LCFE module), with identical training strategies and dataset partitioning across all experiments. Among them, ++LCFE denotes the integration of the LCFE module into the GCFE framework, corresponding to the proposed ACFF module. The experimental results demonstrate that the +GCFE increases an mAP of 71.9%, while ACFF further elevates the mAP to 72.3%—representing a 1.6% improvement over the best comparative method, CMDAF (70.7%), and a 0.4% enhancement over GCFE. This performance gain stems from ACFF's dynamic weight allocation mechanism, which adaptively strengthens complementary feature representations between IR and RGB modalities, particularly demonstrating superior discriminative capability in UAV small object detection scenarios. Regarding resource consumption, the +GCFE module requires 5.7 MB of model parameters and 13.5 GFLOPs of computational complexity. ACFF introduces an additional 0.8 MB of model parameters and 2.4 GFLOPs compared to GCFE, and 2.6 MB of model parameters with 4.3 GFLOPs relative to CMDAF. However, its precision gain per computational unit (ΔmAP/ΔGFLOPs = 0.37%) significantly surpasses methods like DMFF (0.25%), validating the efficacy of computational resource allocation. To ensure comparative fairness, we adapted the original designs of CMDAF [61] and CFT [31] by implementing multimodal feature fusion through element-wise addition, thereby maintaining a unified fusion operation benchmark across all methodologies.

Specifically, ACFF attains an MR of 26.8% and an FAR of 2.8%, which represents a reduction of 1.3% in MR and 0.7% in FAR compared to the GCFE configuration (28.1% MR and 3.5% FAR). These reductions are notably larger

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TABLE VII
ANALYSIS OF THE EFFECTIVENESS OF THE LHA ATTENTION MECHANISM

Method		Params. (MB)	GFLOPs	P (%)	R (%)	mAP (%)
Baseline		6.54	15.9	68.4	70.3	72.3
+SE[37]		6.54	15.9	73.0	67.8	72.4
+CBAM[39]		6.55	15.9	72.6	68.2	72.8
+CA[63]		6.55	15.9	72.1	69.1	72.5
+ ACMix[45]		6.83	16.5	70.6	70.3	73.0
+LHA	w/o CS	6.80	16.4	70.6	71.2	74.6
(**Ours**)	w/ CS	6.80	16.4	71.5	73.1	75.1

than the marginal mAP gain and demonstrate the module's enhanced capability in reducing both missed detections and false positives—a critical advantage in real-world UAV applications where reliability is paramount. The LCFE component, through its localized CFE, contributes precisely to this more refined feature integration, thereby improving detection confidence and spatial accuracy.

3) Impact of the LHA Attention Mechanism on the Model Performance: To validate the effectiveness of the LHA attention mechanism in the task of small target detection by UAVs, we established a baseline based on the dual-branch YOLOv10 architecture and the ACFF fusion module, and then carried out comparative experiments. As presented in Table VII, in contrast to the baseline, after the introduction of the LHA module, the model's parameter count increased by merely 0.26 MB, and the computational complexity increased by 0.5 GFLOPs. However, there was a remarkable enhancement in detection performance: the precision rate (P) increased by 3.1%, the recall rate (R) was elevated by 2.8%, and the mAP increased by 2.8%. It is noteworthy that the performance of LHA in terms of the mAP metric outperformed that of the current mainstream attention mechanisms (SE [37], CBAM [39], CA [63], and ACMix [45]) by 1.5%–2.7%. This outcome fully attests to the superiority of the LHA attention mechanism in capturing small target features and suppressing complex backgrounds. To further verify the impact of the channel shuffle technique on the module's performance, this article conducts ablation experiments on the LHA with and without channel shuffling, where "w/o CS" indicates no use of channel shuffle and "w/ CS" indicates the use of channel shuffle. The experimental results show that the channel shuffle technique does not change the model's parameter quantity or computational cost, but after adopting this technique, P, R, and mAP increase by 0.9%, 1.9%, and 0.5%, respectively. Although channel shuffle does not increase computational cost, it effectively enhances the model's detection capability, further verifying the rationality and effectiveness of the LHA structure design.

E. Visualization Analysis

In order to evaluate the performance of the CA-YOLO algorithm proposed in this study in small target detection tasks in a more intuitive and effective manner, we selected representative detection samples from the DroneVehicle, LLVIP, and FLIR datasets and carried out a visualization analysis on them.

Fig. 7 illustrates the comparative performance of CA-YOLO, C^{2}former, and YOLOv10n on the DroneVehicle dataset. During the daytime, in the detection of RGB and IR images, the YOLOv10n algorithm demonstrates issues with missed detections for car-class targets. Specifically, the missed targets in RGB and IR images are highlighted by blue and red circles, respectively. While C^{2}former also exhibits some missed detections, it shows a marked improvement compared to YOLOv10n. The missed targets in RGB and IR images for C^{2}former are denoted by blue and red circles, respectively. In contrast, CA-YOLO entirely avoids such missed detection problems. At night, due to poor lighting conditions, the YOLOv10n algorithm exhibits significant missed detection phenomena in RGB images (the missed detection areas are indicated by blue circles). In IR images, although the missed detection situation is somewhat alleviated, false detection issues arise. Specifically, the left red circle indicates a missed car, while the right red circle represents a noncar target misclassified as a freight car. For C^{2}former, only a limited number of missed detections are present in the detection results of RGB and IR images (marked by blue and red circles, respectively). Clearly, compared to YOLOv10n, C^{2}former has reduced both the missed detection rate and the false detection rate. It is noteworthy that the CA-YOLO algorithm does not exhibit any missed detections or false detections during nighttime. The experimental results demonstrate that CA-YOLO significantly enhances the robustness of multimodal object detection under varying lighting conditions during both day and night.

Fig. 8 illustrates the comparative results of CA-YOLO, C^{2}former, and YOLOv10n on the LLVIP dataset. During the daytime, in the detection of RGB and IR images, the YOLOv10n algorithm exhibits missed detection issues for "person-class" targets. Specifically, the missed targets in RGB and IR images are marked by blue and red circles, respectively. C^{2}former shows a reduction in the missed detection rate compared to YOLOv10n. The missed targets of C^{2}former in RGB and IR images are also denoted by blue and red circles, respectively. In contrast, CA-YOLO completely circumvents the problem of missed detections. At night, the missed detection rate of YOLOv10n for RGB images is significantly higher than that for IR images. This can be mainly attributed to insufficient lighting at night and the interference of vehicle headlights, which makes it difficult to detect targets. Specifically, the missed targets in RGB and IR images are marked by blue and red circles, respectively. C^{2}former significantly reduces the missed detection rate compared to YOLOv10n. This is due to C^{2}former's integration of the advantageous features of RGB and IR images. It is worth emphasizing that the CA-YOLO algorithm demonstrates excellent performance at night. It not only completely eliminates the problem of missed detections but also achieves high-precision detection with zero false detections. The experimental results further validate that the proposed CA-YOLO algorithm possesses good robustness in the multimodal small target detection task under different lighting conditions during day and night.

Fig. 9 depicts the results of a comparative analysis of CA-YOLO, C^{2}former, and YOLOv10n on the FLIR dataset.

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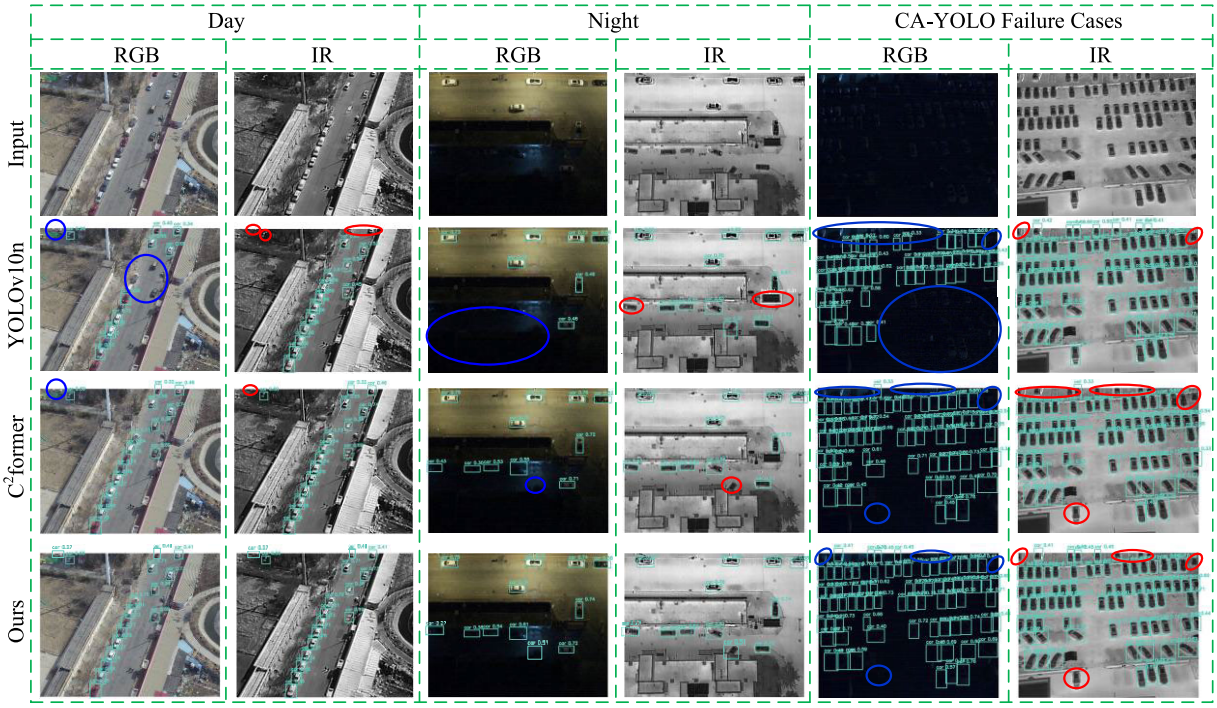


Fig. 7. Comparison results of CA-YOLO, C^{2}former, and YOLOv10n on the DroneVehicle dataset.

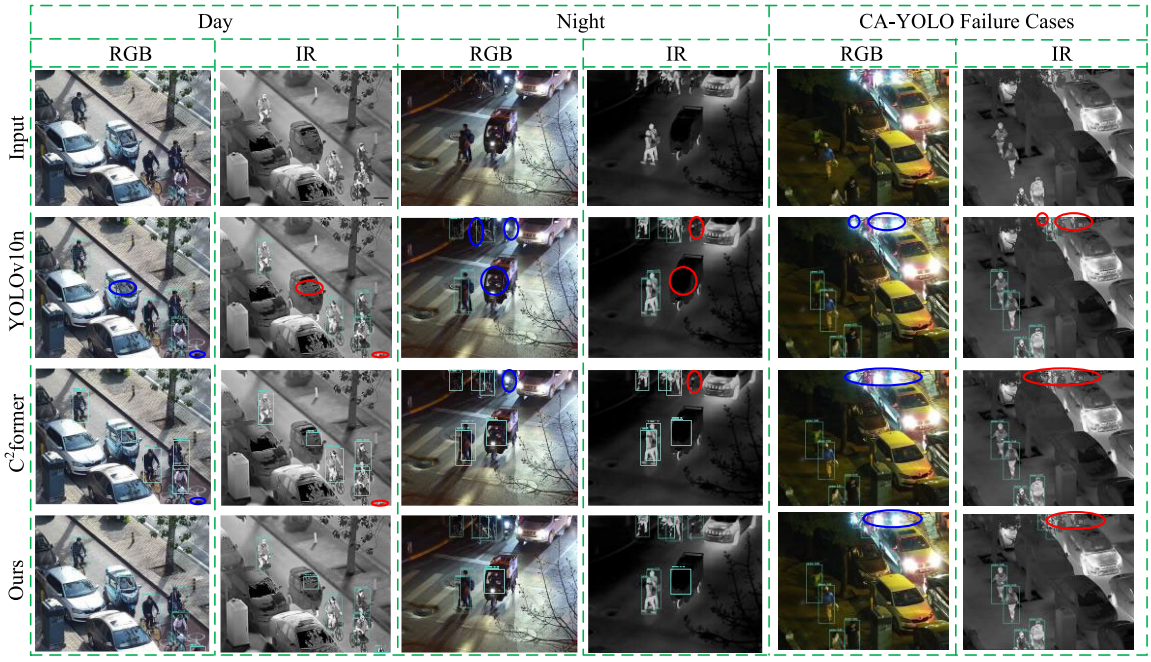


Fig. 8. Comparison results of CA-YOLO, C^{2}former, and YOLOv10n on the LLVIP dataset.

During the daytime, the YOLOv10n algorithm exhibits missed detection issues when processing RGB images. The missed detection targets are indicated by blue circles. For IR images, both missed detection and false detection occur. Specifically, the left-hand red circle represents a false detection where a "sign" is misclassified as a "car," while the right-hand red circle indicates a missed detection of a "person" with small-scale features. In contrast to YOLOv10n, C^{2}former does not present any missed detection or false detection cases. However, its detection accuracy is marginally lower than that of CA-YOLO. At night, YOLOv10n encounters severe missed detection problems during the detection of RGB images. The missed detection areas are marked by blue circles. For IR images, although the missed detection rate is lower than that of RGB images, missed detections still remain. C^{2}former also experiences missed detections, with the missed targets in RGB and IR images marked by blue and red circles, respectively. Nevertheless, compared to YOLOv10n,

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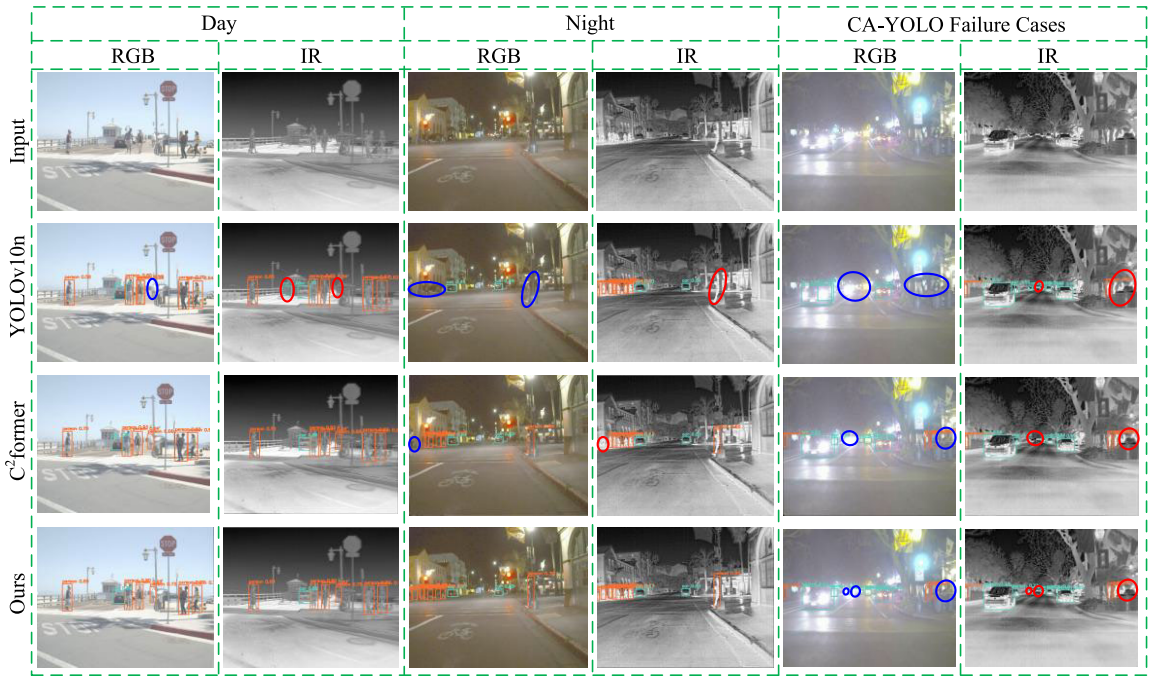


Fig. 9. Comparison results of CA-YOLO, C^{2}former, and YOLOv10n on the FLIR dataset.

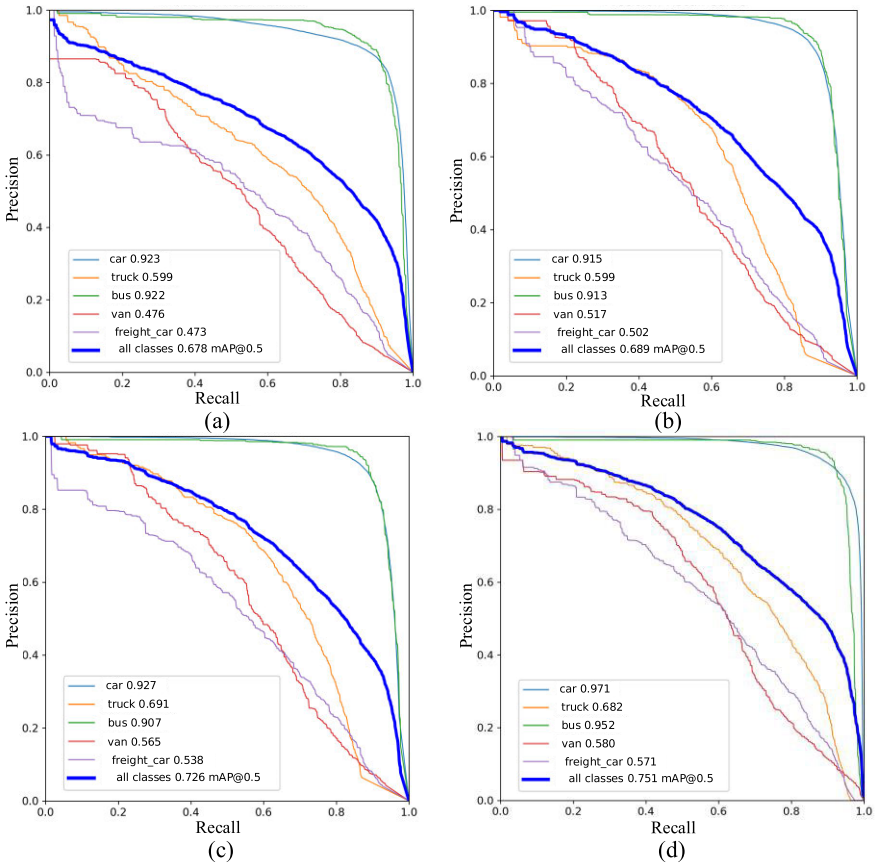


Fig. 10. P–R curves of different models under the DroneVehicle dataset. (a) YOLOv10n—IR. (b) YOLOv10n—RGB. (c) C^{2}former. (d) Ours.

C^{2}former shows a significant improvement, which can be attributed to its unique multimodal fusion mechanism. Notably, CA-YOLO demonstrates no missed detections during the night, showcasing exceptional detection capabilities. This outstanding performance can be ascribed to CA-YOLO's distinctive cross-modal feature fusion and attention mechanism,

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which effectively enhances the accuracy of small target detection. The above visual experimental results offer compelling evidence for the high performance of the CA-YOLO algorithm.

In order to conduct a comprehensive analysis of the dynamic variation patterns of the proposed CA-YOLO algorithm's performance in small target detection, this study undertakes an investigation using the DroneVehicle dataset as the experimental sample. Based on the data presented in Table II, the single-modal baseline model YOLOv10n and the multimodal high-performing model C^2former are selected as control groups for comparative analysis. The P–R curve depicted in Fig. 10 reveals that despite the differences in the mAP among various models, they all demonstrate a consistent tendency of precision decline as the recall rate rises. In the single-modal scenario, the mAP of YOLOv10n in the RGB modality is significantly superior to that in the IR modality. The cross-modal comparison results indicate that by integrating the complementary features of RGB and IR, the multimodal model achieves a 6.2%–7.3% increase in mAP compared to the single-modal model. Furthermore, when compared with C^2former, although CA-YOLO exhibits some local performance discrepancies in the detection of "truck-class" targets, it attains mAP improvements of 4.4%, 1.5%, 4.5%, and 3.3% in the detection of "car," "van," "bus," and "freight car" targets, respectively. This showcases the algorithm's exceptional comprehensive detection capabilities. The above experimental findings validate the effectiveness and robustness of the proposed algorithm in the task of small target detection for UAVs.

To provide a more comprehensive and objective performance evaluation, we have supplemented Figs. 7–9 with failure cases of CA-YOLO under some extreme scenarios.

Fig. 7 adds an example of missed detection in high-density vehicle environments at night, where targets are mutually occluded. The background is complex, causing some edge vehicles to be overlooked due to high feature similarity. Fig. 8 adds cases where pedestrian targets are missed under strong direct light or local overexposure; strong light interference blurs target contours, affecting effective inter-modal feature complementarity. Fig. 9 adds failure cases for extremely small-scale targets and strong reflection/occlusion scenes, where targets occupy extremely low pixel areas and lack texture information, making accurate localization difficult for the model. These failure cases mostly occur in challenging environments with abnormal lighting, extremely dense targets, or very small scales, reflecting the limitations of the current method in feature representation and cross-modal fusion. Nevertheless, CA-YOLO still demonstrates significant performance advantages in most common scenarios, and its adaptive fusion and attention mechanisms show clear value in improving detection robustness.

V. Conclusion

This article presents a CA-YOLO framework based on the ACFF and attention mechanism for small object detection in UAV remote sensing images. The framework utilizes a dual-branch backbone network to extract complementary features from RGB and IR modalities. An ACFF module is designed to dynamically integrate global semantic and local detail information. In addition, an LHA is introduced to enhance small object features while suppressing background noise. Extensive experiments on multiple public datasets demonstrate that the proposed method achieves high detection accuracy and shows advantages in reducing MR and false positive rate. Although the ACFF and LHA modules are designed with lightweight considerations, the dual-branch structure and cross-modal interaction mechanism introduce additional computational overhead, which may affect real-time inference performance on resource-constrained embedded UAV platforms. Future work will explore more efficient network architecture designs or model compression techniques to further improve inference speed while maintaining detection accuracy.

Appendix

This Appendix supplements the detailed implementation details of the GCFE and LCFE modules in the ACFF module.

A. Detailed Implementation of the GCFE Module

First, for the given RGB input feature map $F_{RGB} \in R^{H\times W\times C}$ and IR input feature map $F_{IR} \in R^{H\times W\times C}$, we apply hybrid pooling (modified average and max pooling) for spatial compression, avoiding information loss and high computation. A learnable factor α adaptively fuses the pooled features

$$\begin{cases}\tilde{F}^1_{RGB} = Avgpooling(F_{RGB})\\ \tilde{F}^2_{RGB} = Maxpooling(F_{RGB})\end{cases}$$
$$\begin{cases}\tilde{F}^1_{IR} = Avgpooling(F_{IR})\\ \tilde{F}^2_{IR} = Maxpooling(F_{IR})\end{cases} \tag{22}$$
$$T_{RGB} = \alpha\tilde{F}^1_{RGB} + (1-\alpha)\,\tilde{F}^2_{RGB}$$
$$T_{IR} = \alpha\tilde{F}^1_{IR} + (1-\alpha)\,\tilde{F}^2_{IR} \tag{23}$$

where $\tilde{F}^1_{RGB}$ and $\tilde{F}^2_{RGB} \in R^{(H\times W)\times C}$ are compressed features from F_{RGB}, and similarly for F_{IR}. $\alpha \in [0,1]$ is learnable. $T_{RGB} \in R^{(H\times W)\times C}$ represents the feature result obtained after the adaptive fusion of $\tilde{F}^1_{RGB}$ and $\tilde{F}^2_{RGB}$. Similarly, $T_{IR} \in R^{HW\times C}$.

Second, we project T_{RGB} to obtain the value matrix $V_R \in R^{HW\times C}$, and the key matrix $K_R \in R^{HW\times C}$ and T_{IR} to obtain query matrix $Q_I \in R^{HW\times C}$

$$V_R = T_{RGB}\cdot W^V, \quad K_R = T_{RGB}\cdot W^K, \quad Q_I = T_{IR}\cdot W^Q \tag{24}$$

where W^V, W^K, and $W^Q \in R^{HW\times C}$ are the weight matrices.

Third, we project the cross-modal attention vector Z_R, and this is combined with T_{RGB} to get $T'_{RGB} \in R^{HW\times C}$

$$Z_R = \text{soft max}\left(\frac{Q_I K_R^T}{\sqrt{D_K}}\right)\cdot V_R \tag{25}$$
$$T'_{RGB} = T_{RGB} + Z_R\cdot W^O \tag{26}$$

where K_R^T is the transpose of the K_R matrix from the RGB modality, $\sqrt{D_K}$ is the scale factor, and $W^O \in R^{HW\times C}$ represents the weight matrix.

Finally, T'_{RGB} is enhanced by an FFN and reshaped to $T^2_{RGB} \in R^{H\times W\times C}$

$$T^1_{RGB} = T'_{RGB} + FFN\left(T'_{RGB}\right) \tag{27}$$

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$$T_{RGB}^{1} \in R^{HW\times C} \Rightarrow T_{RGB}^{2} \in R^{H\times W\times C}. \tag{28}$$

Similarly, the eigenvector obtained at the end of the IR modality is $T_{IR}^{2} \in R^{H\times W\times C}$.

B. Detailed Implementation of the LCFE Module

First, we generate a box-level mask from bounding box annotations: background = 0 and target = 1. For the RGB feature map F_{RGB}, the mask $L_{RGB} \in R^{H\times W\times 1}$ is predicted via three convolutional layers

$$L_{RGB} = \sigma\left[Conv_{1\times 1}\left(Conv_{3\times 3}\left(Conv_{3\times 3}\left(F_{RGB}\right)\right)\right)\right] \tag{29}$$

where σ is the sigmoid, while $Conv_{1\times 1}$ and $Conv_{3\times 3}$ correspond to convolutional layers with kernel sizes of 1 and 3, respectively.

Second, we then compute cosine similarity between L_{RGB} and F_{RGB} to get the correlation vector $L_{RGB}^{1} \in R^{1\times 1\times C}$

$$L_{RGB}^{1} = CoS\left(L_{RGB}, F_{RGB}\right) \tag{30}$$

where L_{RGB}^{1} denotes the correlation between the box-level mask L_{RGB} and the feature map F_{RGB}. CoS denotes the cosine similarity function.

Third, an MLP enhances L_{RGB}^{1} to $L_{RGB}^{2} \in R^{1\times 1\times C}$

$$L_{RGB}^{2} = \sigma\psi\left(L_{RGB}^{1}\right) \tag{31}$$

$$\psi\left(L_{RGB}^{1}\right) = Conv_{1\times 1}\left(Conv_{1\times 1}\left(L_{RGB}^{1}\right)\right) \tag{32}$$

where ψ denotes the MLP function.

Fourth, we multiply L_{RGB}^{2} with F_{RGB} to enhance target regions. The enhanced feature $L_{RGB}^{3} \in R^{H\times W\times C}$ is derived

$$L_{RGB}^{3} = L_{RGB}^{2} \otimes F_{RGB}. \tag{33}$$

Finally, a spatial gating module is applied: average and max pooling of L_{RGB}^{3} are concatenated, processed by a 7×7 Conv, and multiplied with L_{RGB}^{3} to output $L_{RGB}^{4} \in R^{H\times W\times C}$

$$\begin{aligned} L_{RGB}^{4} &= Conv_{7\times 7}\left(Cat\left(AvgPool\left(L_{RGB}^{3}\right), MaxPool\left(L_{RGB}^{3}\right)\right)\right) \\ &\quad \otimes L_{RGB}^{3}. \end{aligned} \tag{34}$$

Similarly, $L_{IR}^{4} \in R^{H\times W\times C}$ is obtained for the IR branch.

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